# IO-Link Data Reference Guide: CL50 Pro

## IO-Link Data Map

This document refers to the following IODD file: Banner_Engineering-CL50RGBIOL-20210903-IODD1.1.xml. The IODD file and support files can be found on www.bannerengineering.com under the download section of the product family page.

## Communication Parameters

The following communication parameters are used.

| Parameter | Value |
|---|---|
| IO-Link revision | V1.1 |
| Process Data In length | N/A |
| Process Data Out length | 3-bytes |
| Bit Rate | 38400 bps |
| Minimum cycle time | 20 ms |

| Parameter | Value |
|---|---|
| Port class | A |
| SIO mode | No |
| Smart sensor profile | N/A |
| Block parameterization | Yes |
| Data Storage | Yes |

## IO-Link Process Data In (Device to Master)

Not applicable.

## IO-Link Process Data Out (Master to Device)

| Run Mode | | | |
|---|---|---|---|
| **Subindex** | **Name** | **Number of Bits** | **Data Values** |
| 1 | Color 1 | 4 | 0 = Green, 1 = Red, 2 = Orange, 3 = Amber, 4 = Yellow, 5 = Lime Green, 6 = Spring Green, 7 = Cyan, 8 = Sky Blue, 9 = Blue, 10 = Violet, 11 = Magenta, 12 = Rose, 13 = White, 14 = Custom 1, 15 = Custom 2 |
| 2 | Color 2 | 4 | 0 = Green, 1 = Red, 2 = Orange, 3 = Amber, 4 = Yellow, 5 = Lime Green, 6 = Spring Green, 7 = Cyan, 8 = Sky Blue, 9 = Blue, 10 = Violet, 11 = Magenta, 12 = Rose, 13 = White, 14 = Custom 1, 15 = Custom 2 |
| 3 | Animation | 3 | 0 = Off, 1 = Steady, 2 = Flash, 3 = Two Color Flash, 4 = Intensity Sweep |
| 4 | Pulse Pattern | 3 | 0 = Normal, 1 = Strobe, 2 = Three Pulse, 3 = SOS, 4 = Random |
| 5 | Speed | 2 | 0 = Medium, 1 = Fast, 2 = Slow |
| 6 | Color 1 Intensity | 3 | 0 = High, 1 = Low, 2 = Medium, 3 = Off, 4 = Custom |
| 7 | Color 2 Intensity | 3 | 0 = High, 1 = Low, 2 = Medium, 3 = Off, 4 = Custom |
| 8 | Audible State | 2 | 0 = Off, 1 = On, 2 = Pulsed, 3 = SOS Pulse |

| **Octet 0** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Bit Offset | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Subindex | 8 | | 7 | | | 6 | | |
| Value | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Example | Audible State: On | | Color 2 Intensity: High | | | Color 1 Intensity: High | | |

| **Octet 1** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Bit offset | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| Subindex | 5 | | 4 | | | 3 | | |
| Value | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| Example | Speed: Fast | | Pulse Pattern: Normal | | | Animation: Two Color Flash | | |

| **Octet 2** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Bit offset | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Subindex | 2 | | | | 1 | | | |
| Value | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| Example | Color 2: Red | | | | Color 1: Green | | | |

## Parameters Set Using IO-Link

These parameters can be read from and/or written to an IO-Link model of the CL50 Pro.

| Index | Subindex | Name | Length | Value Range | Default | Access Rights | Data Storage? | AOI |
|---|---|---|---|---|---|---|---|---|
| 0 | 1-16 | Direct Parameter Page 1 (incl. Vendor ID & Device ID) | | | | rw | | |
| 1 | 1-16 | Direct Parameters Page 2 | | | | rw | | |
| 2 | | Standard Command | | 130 = Restore Factory Settings | | wo | | |
| 3-11 | | | | | | | | |
| 12 | | **Device Access Locks** | | | | | | |
| | 0 | Parameter (write) Access Lock | Boolean | 0 = off, 1 = on | 0 | rw | y | |
| | 1 | Data Storage Lock | Boolean | 0 = off, 1 = on | 0 | rw | y | |
| | 2 | Local Parameterization Lock | Boolean | 0 = off, 1 = on | | rw | y | |
| | 3 | Local User Interface Lock | Boolean | 0 = off, 1 = on | | rw | y | |
| 13-15 | | | | | | | | |
| 16 | | Vendor Name string | | Banner Engineering Corporation | | ro | | |
| 17 | | Vendor Text string | | More Sensors. More Solutions. | | ro | | |
| 18 | | Product Name string | | CL50 | | ro | | |
| 19 | | Product ID string | | CL50P*K*Q | | ro | | |
| 20 | | Product Text string | | CL50 Pro | | ro | | |
| 21 | | Serial Number | | | | ro | | |
| 22 | | Hardware Revision | | | | ro | | |
| 23 | | Firmware Version | | | | ro | | |
| 24 | | App Specific Tag (user defined) | | | | rw | y | |
| 25-35 | | | | | | | | |
| 36 | | Device Status | 8-bit uinteger | 0 = Device is OK, 1 = Maintenance required, 2 = Out of specification, 3 = Functional check, 4 = Failure, 5..255 = Reserved | | ro | | |
| 37 | 1-6 | Detailed Device Status | Array[6] of 3-octet | | | ro | | |
| 38-63 | | | | | | | | |
| 75 | | **Additional Settings** (subindex access supported) | | | | | | |
| | 1 | Custom Intensity (0 - 100%) | 8-bit uinteger | 0-100 | 100 | rw | y | |
| | 2 | Custom Flash Rate (0.5 - 20) | 8-bit uinteger | 0.5-20 | 15 | rw | y | |
| | 4 | Restrict To Gamut | Boolean | 0=Off, 1=On | 0 | rw | y | |
| | 5 | Reserved | 7-bit uinteger | | 0 | | | |
| 76 | | **Custom 1** (subindex access supported) | | | | | | |
| | 1 | Red | 8-bit uinteger | 0-255 | 255 | rw | y | |
| | 2 | Green | 8-bit uinteger | 0-255 | 255 | rw | y | |
| | 3 | Blue | 8-bit uinteger | 0-255 | 255 | rw | y | |
| 77 | | **Custom 2** (subindex access supported) | | | | | | |
| | 1 | Red | 8-bit uinteger | 0-255 | 255 | rw | y | |
| | 2 | Green | 8-bit uinteger | 0-255 | 255 | rw | y | |
| | 3 | Blue | 8-bit uinteger | 0-255 | 255 | rw | y | |

## IO-Link Events

Events and Error Types are acyclic transmissions from the IO-Link device to the IO-Link master. Events can be error messages and/or warning or maintenance data.

| Event Types | | |
|---|---|---|
| **Code** | **Type** | **Description** |
| 0 (0x0000) | Notification | No malfunction |
| 20480 (0x5000) | Error | Device hardware fault/Device exchange |
| 20753 (0x5111) | Error | Primary supply voltage under-run/Check tolerance of power supply |

| Error Types | | | |
|---|---|---|---|
| **Code** | **Additional Code** | **Name** | **Description** |
| 128 (0x80) | 0 (0x00) | Device application error - no details | Service has been refused by the device application and no detailed information of the incident is available |
| | 17 (0x11) | Index not available | Access occurs to a not existing device |
| | 18 (0x12) | Subindex not available | Access occurs to a not existing subindex |
| | 32 (0x20) | Service temporarily not available | Parameter is not accessible because of the current state of the device application |
| | 35 (0x23) | Access denied | Write access on a read-only parameter |
| | 48 (0x30) | Parameter value out of range | Written parameter value is outside its permitted value range |
| | 49 (0x31) | Parameter value above limit | Written parameter value is above its specific value limit |
| | 51 (0x33) | Parameter length overrun | Written parameter length is above its predefined length |
| | 52 (0x34) | Parameter length underrun | Written parameter length is below its predefined length |
| | 53 (0x35) | Function not available | Written command is not supported by the device application |
| | 54 (0x36) | Function temporarily unavailable | Written command is not available because of the current state of the device application |
| | 65 (0x41) | Inconsistent parameter set | Parameter inconsistencies were found at the end of the block parameter transfer, device plausibility check failed |



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